Arista Access Node WFD-ACC-48 Whitefield POP Routing Protocol Flap SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4091

Description

The failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Arista Access Node WFD-ACC-48 at Whitefield POP can be attributed to various underlying causes, including misconfiguration of BGP neighbor relationships, incorrect OSPF neighbor authentication parameters, or faulty ISIS LSP database synchronization. Additionally, hardware-related issues such as faulty interface cards or malfunctioning switching fabrics can also contribute to this failure domain. Furthermore, environmental factors like high temperatures, power outages, or electromagnetic interference can also impact the Arista hardware's ability to maintain stable routing protocols. In this SOP, we will diagnose and resolve the root cause of the Routing Protocol Flap failure on WFD-ACC-48, adhering to the Arista CLI dialect.

Pre-requisite Checks

- pre_requisite: Verify the redundant link status between WFD-ACC-48 and the adjacent nodes.

- pre_requisite: Confirm the dispatch of remote hands to the Whitefield POP for on-site assistance, if necessary.

- pre_requisite: Ensure that the Arista hardware is properly powered and cooled.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor relationships	High
Incorrect OSPF neighbor authentication parameters	Medium
Faulty ISIS LSP database synchronization	Medium
Faulty interface cards or malfunctioning switching fabrics	Low
Environmental factors (high temperatures, power outages, or electromagnetic interference)	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP table version 10 BGP table version 10 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path *> * 192.168.1.0/24 10.1.1.1 0 100 32768 i *> * 192.168.2.0/24 10.2.2.2 0 100 32768 i, Description: The show bgp summary command provides a summary of the BGP table, including the BGP table version, status codes, origin codes, and network information. This command helps identify any BGP neighbor relationship misconfigurations.
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Command: show ospf neighbor, Expected Output: OSPFv2 Neighbor Status Neighbor ID Pri State Up Time Addresses 10.1.1.1 1.0 FULL 00:00:00 10.1.1.1/32 10.2.2.2 1.0 FULL 00:00:00 10.2.2.2/32, Description: The show ospf neighbor command displays the OSPF neighbor status, including neighbor ID, priority, state, up time, and addresses. This command helps identify any OSPF neighbor authentication parameter misconfigurations.

Command: show isis database, Expected Output: ISIS LSP Database LSP ID Sequence Age Checksum Sequence Number LSP Size 10.1.1.1.10.1.1 1000 00:00:00 0x12345678 100 64 10.2.2.2.10.2.2 1000 00:00:00 0x12345678 100 64, Description: The show isis database command displays the ISIS LSP database, including LSP ID, sequence, age, checksum, sequence number, and LSP size. This command helps identify any ISIS LSP database synchronization issues.
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Resolution Steps

Step 1: Step: Verify BGP neighbor relationships, Command: show bgp summary, Description: Verify the BGP table and identify any misconfigured BGP neighbor relationships.

Step 2: Step: Correct OSPF neighbor authentication parameters, Command: configure terminal ospf authentication-key , Description: Correct any incorrect OSPF neighbor authentication parameters using the configure terminal command and the ospf authentication-key command.

Step 3: Step: Synchronize ISIS LSP database, Command: isis database sync, Description: Synchronize the ISIS LSP database using the isis database sync command.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version 10 BGP table version 10 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path *> * 192.168.1.0/24 10.1.1.1 0 100 32768 i *> * 192.168.2.0/24 10.2.2.2 0 100 32768 i, Description: Verify that the BGP table is stable and that the BGP neighbor relationships are correct.

Command: show ospf neighbor, Expected Output: OSPFv2 Neighbor Status Neighbor ID Pri State Up Time Addresses 10.1.1.1 1.0 FULL 00:00:00 10.1.1.1/32 10.2.2.2 1.0 FULL 00:00:00 10.2.2.2/32, Description: Verify that the OSPF neighbor status is stable and that the OSPF neighbor authentication parameters are correct.

Command: show isis database, Expected Output: ISIS LSP Database LSP ID Sequence Age Checksum Sequence Number LSP Size 10.1.1.1.10.1.1 1000 00:00:00 0x12345678 100 64 10.2.2.2.10.2.2 1000 00:00:00 0x12345678 100 64, Description: Verify that the ISIS LSP database is stable and that the ISIS LSP database synchronization is correct.
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Roll-back Procedures

- Step: Revert BGP neighbor relationships to previous state, Command: configure terminal undo bgp neighbor , Description: Revert any changes made to the BGP neighbor relationships using the configure terminal command and the undo bgp neighbor command.

- Step: Revert OSPF neighbor authentication parameters to previous state, Command: configure terminal undo ospf authentication-key , Description: Revert any changes made to the OSPF neighbor authentication parameters using the configure terminal command and the undo ospf authentication-key command.

- Step: Revert ISIS LSP database synchronization to previous state, Command: undo isis database sync, Description: Revert any changes made to the ISIS LSP database synchronization using the undo

isis database sync command.